SE(3)-WAM: A Design Study of History, Future Representation, and World–Action Interaction for Dynamic Manipulation

Anonymous CVPR submission

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Study draft: results and final model selection pending.

Abstract

Dynamic manipulation requires actions to remain useful while the scene changes between observation and execution. A longer visual history, a more detailed future prediction, or a more tightly coupled world–action model may each help, but also consume the latency and computation needed to act. We organize this problem as a controlled design study under common observation, training, and execution budgets. Q1 compares how finite history represents motion and occlusion. Q2 compares future variables and prediction granularity, including target-only, context-only, uniform, and heterogeneous representations. Q3 compares backbone organization and information flow, separating auxiliary supervision, online forecast consumption, and joint generation. A dynamic benchmark connects these choices to contact timing, relative pose, success, safety, and end-to-end cost. Selected cross-comparisons test whether design rankings survive changes in the other axes. SE(3)-WAM denotes the instance selected by this study, rather than a predetermined combination of $\mathrm{SE}(3)$*, flow, and a one-way bridge. The comparative findings, final configuration, and external validation remain **[pending]**; no numerical improvement is claimed in this draft.*

1. Introduction

A robot acts on a scene that has already changed. Camera exposure, transfer, inference, queuing, and actuation separate an observation from the command's physical effect. During that interval, a target may translate or rotate, an obstacle may close a corridor, or contact may change the motion. Faster inference and continuous execution reduce observation age [11, 14]; they do not make a single image reveal velocity or make all future visual detail equally relevant to an action. Figure 1 illustrates the resulting decision at execution time.

Existing approaches make different choices along this chain. Historical motion and future supervision support dynamic policies [4]; predicted visual features can become an online action condition [10, 19]. Geometric world models expose metric motion [5, 17], whereas auxiliary world prediction can help training without requiring test-time imagination [16]. These choices change the available information, the training signal, and the computation before execution. A gain observed after changing all three cannot identify which information or interaction was useful.

We therefore ask: *under matched observation-to-execution delay, capacity, and inference budgets, how should a policy encode history, represent the future, and connect world modeling to action?* The object of the study is an action-level decision, not prediction fidelity alone. A forecast may reconstruct motion accurately but arrive too late, omit a relevant obstacle, or never affect the chosen action. Conversely, a compact representation may suffice when the task depends on only a contact point or a time window.

The study has three formal design questions. **Q1** compares finite-history encodings, including their time, coordinate, and visibility information. **Q2** compares predicted variables and granularity: what belongs to the target, what belongs to task-relevant context, and what detail each requires. **Q3** compares backbone organization and world–action information flow. Auxiliary supervision, online consumption, and joint generation are distinct mechanisms within that comparison. The hypothesis that the target needs finer future detail than its context is tested, not imposed as a universal rule.

We first establish a common protocol and a reference system, then compare Q1, Q2, and Q3 while stating the reference choices held fixed. Selected cross-checks revisit history after changing the future representation and revisit that representation after changing information flow. SE(3)-WAM names the model selected from these comparisons. Target $\mathrm{SE}(3)$, dense flow, compact summaries, and a one-way bridge remain possible components until evidence supports their use. Useful negative results may narrow an instance without removing a design question from the study.

The manuscript is organized around three intended contributions:

- A physical-time dynamic-manipulation protocol with ex-

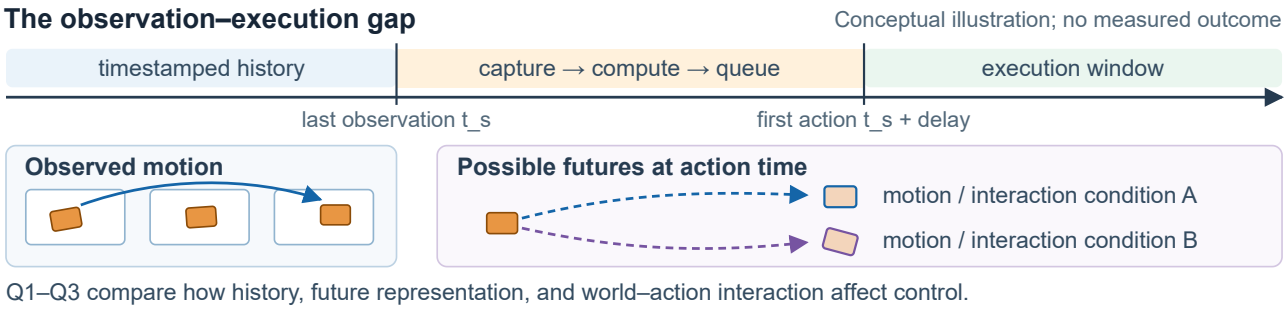


Figure 1. The action-time decision problem, shown conceptually rather than as a measured rollout. An observation and the command's effect occur at different physical times. History can disambiguate motion, a future representation can retain action-relevant state, and world–action interaction determines how that information reaches control. No illustrated outcome establishes a performance advantage.

plicit information boundaries, paired conditions, and action-level evaluation.

- A common-budget study of history encoding, future representation and granularity, and world–action organization, including their selected interactions.
- A study-selected SE(3)-WAM instance, evaluated through the same action, prediction, safety, and cost measures rather than through prediction error alone.

The result slots in Sec. 5 remain pending. Full protocols, candidate mathematics, configuration records, and extended result slots are in the independent Supplementary Material, Secs. A–G.

2. Related Work

History and dynamic execution. Reachability-aware dynamic grasping [1] and conveyor manipulation [7] connect motion to interception. DynamicVLA and VLASH address changing scenes through continuous or asynchronous execution [11, 14]; PUMA combines historical motion with future-feature supervision [4]. Our Q1 comparison concerns what information a bounded history retains, with execution behavior and observation age reported alongside the encoding. It does not equate more frames with a controlled motion signal.

Future variables and control. DBC-TFP and AHEAD expose predicted visual information to action [10, 19]; LaWAM and FutureRTC consider latent futures and execution alignment [2, 6]. PhysMani, LaMP, and PointWorld offer different geometric motion interfaces [5, 13, 17]. Action-conditioned world evolution, exogenous motion, and a desired action flow [21] are different prediction targets. Q2 compares their action-relevant roles and granularity under stated input and budget conditions, rather than declaring one representation best across these settings.

World–action organization. World supervision and online prediction need not have the same benefit: Fast-WAM studies the need for test-time imagination [16], while representation alignment and object-addressable interfaces motivate examining how futures influence actions [8, 9]. OpenWAM directly compares VTA, ATV, Joint, and Decoupled programs on a shared robot-video foundation and Mixture-of-Transformers architecture [15]. Q3 therefore examines conditional choices rather than claiming the first information-flow comparison: execution delay and budget, and interactions with history and future representation, define our intended scope. Component interventions interpret the resulting system comparisons. ACT, Diffusion Policy, and DP3 provide action-model references [3, 18, 20]. Holding their relevant action objective, input, and execution conditions common prevents a decoder change from being attributed to a future representation. None of these precedents makes history, SE(3), or a world–action bridge alone a new contribution; our intended evidence is the controlled design comparison and its action-level explanation.

3. Common Problem and Evaluation Protocol

Physical time and available information. At policy query t, the latest sensor capture time is τ_t^s. A method builds history $\mathcal{H}_t$ from available timestamped observations and robot state; its context, validity mask, and action memory are method-side choices. The target for action index h is evaluated at physical time

$$\tau_h^\star = \tau_t^s + \ell_t + h\Delta_a, \quad h = 0, \dots, H-1, \qquad (1)$$

where ℓ_t is the observation-to-application delay and Δ_a is the high-level action interval. Fixed delay is a controlled episode condition; measured runtime delay is a separate systems quantity. A frame index alone does not define either. The release, rates, horizons, and runtime used for the final comparisons remain **[protocol binding pending]**; historical settings in Supplementary Sec. A are candidate settings, not a completed evaluation.

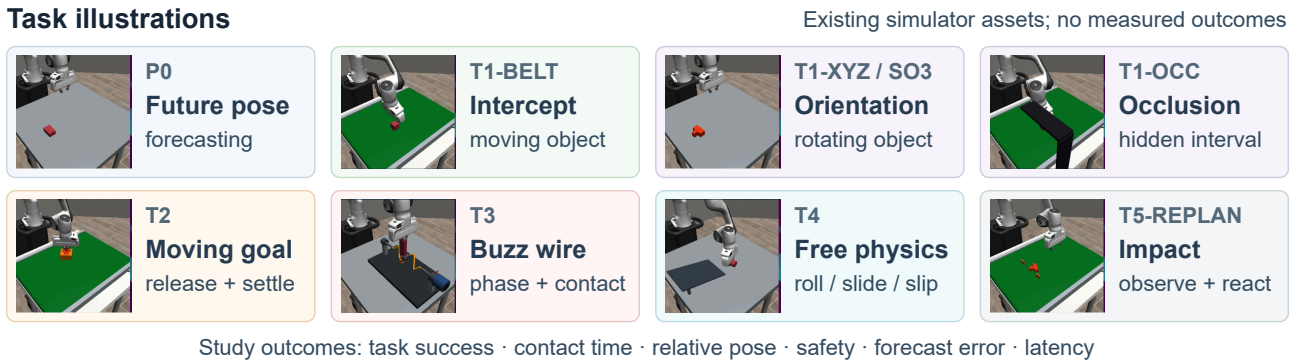


Figure 2. Task illustrations from the existing benchmark assets. Controlled factors cover target translation and rotation, occlusion, moving placement, constrained passage, free motion, and post-contact adaptation. The images identify task conditions, not measured method outcomes; the final task release and evaluation records remain pending.

Benchmark and information boundary. The DB0 diagnostic layer uses rigid-object dynamic manipulation to separate motion observability, contact geometry, timing, and context. Its task families include moving-target grasping, rotating targets, physical occlusion, moving-goal placement, constrained traversal, free dynamics, and replanning after impact (Fig. 2). Simulation uses MuJoCo and robosuite [12, 22]. Standard visual policies receive RGB-D, calibration, timestamps, and robot state; target truth, scene identities, contacts, future labels, and teacher records remain supervision or evaluator information. Object-GT inputs are separately labeled information-oracle studies, not evidence of visual localization. A baseline retaining end-effector history is not strictly memoryless even if its image input is current-only.

Pairing, splits, and outcomes. Exact-current counterfactual groups hold current appearance and physical state fixed while histories imply different futures. Group descendants stay in one split, and methods share paired test conditions. Direction ambiguity, occlusion, and sampling intervals test whether historical information is necessary; necessity is not inferred from aggregate success or a decodable feature probe. Success requires an ordered physical task outcome, rather than incidental touch or a final pose alone. Grasping must establish valid grasp, clearance, and stable hold; moving placement must satisfy containment and release; constrained passage must respect collision conditions. Full event definitions and final thresholds belong in Supplementary Sec. A. Unannounced events after a query test adaptation, not deterministic prediction of unobservable events. Post-contact world targets are valid only under matched applied actions, or are masked when that future becomes action-dependent.

Shared budgets and metrics. Comparisons state observations and history support, training data and exposure, capacity, action interface, controller, prediction horizon, and inference budget. The common execution convention drops stale commands and accepts temporally valid replacements; its effect must not be attributed to representation design. Primary action measures are paired task success, task completion, contact-time and relative-pose error, with safety failures reported separately. Representation errors retain their own physical units and valid support. Cost includes batch-one end-to-end latency, action age, and memory, not only decoder FLOPs. Individual success rates use binomial intervals; differences preserve paired groups and seed structure. Final sample counts and uncertainty estimates remain **[pending]**. Test-OOD, native external benchmarks, static compatibility, and real-robot outcomes remain separate evaluation tracks.

Contact-level readouts. Prediction and action evaluation share physical time but measure different objects. A pose forecast is compared with the object's state at $\tau_h^\star$; contact error is evaluated at the rollout's actual contact event, not at the forecast frame with the lowest error. For a task with a declared reference contact time $t_c^\star$, the candidate timing readout is $e_c^t = |t_c - t_c^\star|$. The reference annotation, treatment of an admissible contact window, and valid event population remain **[contact-metric protocol]**. They must be common to all policies. For a declared desired object-to-end-effector transform $\boldsymbol{G}$, the corresponding relative-pose residual is

$$\boldsymbol{E}_c = \boldsymbol{G}^{-1}\left(\boldsymbol{T}_O^W(t_c)\right)^{-1}\boldsymbol{T}_E^W(t_c). \tag{2}$$

Its translation norm and rotation geodesic retain distinct units. This readout applies only to tasks with that contact reference; placement and traversal use their own declared events. A rollout with no valid contact remains an action failure, rather than receiving zero timing or pose error. Conditional contact and completion statistics report their denominator alongside unconditional success and safety. This prevents a method

from appearing more precise merely by succeeding on fewer, easier episodes.

4. A Controlled World–Action Design Space

The study varies an encoder E_α, a future representation Z_β, and a world–action organization I_γ. Their configuration is $c = (\alpha, \beta, \gamma)$, with a shared observation and action interface. Figure 3 separates these design axes from the common physical-time protocol. The initial reference configuration provides a comparison base; it is not a final architecture selection.

4.1. Q1: Encoding Finite History

The encoder maps a finite support of observations to scene and motion tokens: $\boldsymbol{X}_t = E_\alpha(\mathcal{H}_t)$. Candidate encodings include RGB history, time-aware flow, sparse point trajectories, object or scene states, and recurrent geometric memory. Each specifies coordinate frame, sampling interval, camera motion, visibility, and early-history validity. Raw displacement without elapsed time is not a velocity representation. Methods either share the available history support and token budget or explicitly report the difference. Object pose and kinematics provided by GT are separate information conditions. Q1 is evaluated with a stated reference future target and interaction; its ranking is revisited after Q2 changes what history must support.

An encoding comparison separates access to information from its organization. One contrast keeps the physical history window fixed while changing the token representation; another varies the window within the same encoder. They answer different questions and are not pooled. Flow and trajectories are expressed relative to declared frames with their true time intervals. Camera-induced motion must be accounted for or explicitly retained as part of the input. An early episode with fewer valid observations does not gain synthetic evidence by repeating its reset frame. A current-object-GT control and a history-object-GT control hold the remaining robot-state history common, so that their contrast concerns object information rather than an undeclared change in proprioception.

4.2. Q2: Future Variables and Granularity

Given the same available history and execution times, a future predictor forms

$$\hat{Z}_{\beta,t} = F_\beta(\boldsymbol{X}_t, \{\tau_h^\star\}_{h=0}^{H-1}). \tag{3}$$

The target may be rigid pose, trajectory or contact geometry; task-relevant context may include obstacles, support, moving receptacles, and visibility. Context is defined by its possible effect on contact, reachability, or collision, not by a fixed foreground/background split. Static geometry may matter, whereas unrelated motion need not be predicted.

The comparison distinguishes no future, target-only, context-only, uniform granularity, and heterogeneous granularity. Representation families such as RGB or latent state, point trajectories, scene flow, and target $\mathrm{SE}(3)$ are candidates within these roles. A granularity comparison states its region allocation, spatial and temporal resolution, and decoded-token budget. Changing both the representation and its budget without a matched control cannot identify the value of granularity. The action-relevant hypothesis is that preserving target contact geometry and sufficient contextual constraints may outperform uniform detail at the same cost; the study also allows the uniform alternative or auxiliary-only prediction to prevail.

Role and representation are also distinct factors. Comparing target $\mathrm{SE}(3)$ with whole-scene RGB changes both what is described and how it is described. We therefore record each candidate's target identity source, contextual support, state variables, spatial allocation, temporal sampling, and prediction budget. Where the same representation permits it, target-only, context-only, and combined conditions isolate role coverage. Within a compatible combined representation, uniform and heterogeneous allocations test granularity while retaining the same information source and prediction-token allowance. Cross-family comparisons then report both native prediction error and common action outcomes; a latent reconstruction loss is not numerically ranked against centimeters of pose error. These conditional contrasts support a narrower, interpretable conclusion than a single table in which every row changes role, modality, and cost together.

Temporal resolution is part of granularity. Sampling many future steps at equal frame intervals can allocate most computation away from the command's execution window. Each predictor therefore declares the physical horizons it covers and how the action program queries or pools them. Any extra horizon, adaptive region selection, or forecast refinement is charged to the same end-to-end budget. The study can retain a simpler forecast when additional detail is accurate but does not improve the relevant contact or constraint decision.

For the rigid-target candidate, for example,

$$\hat{\boldsymbol{T}}_h = \mathrm{Exp}(\hat{\boldsymbol{\xi}}_h^\wedge)\hat{\boldsymbol{T}}_t, \qquad \hat{\boldsymbol{T}}_h \in \mathrm{SE}(3). \tag{4}$$

This is an output parameterization, not an equivariance claim. Dense displacement or uncertainty can be added only with its own consumer and evaluation. Predicted external motion, action-conditioned consequences, and desired task trajectories are labeled separately. Supplementary Sec. B gives the inherited $\mathrm{SE}(3)$/flow candidate mathematics; it does not establish that those heads should survive model selection.

4.3. Q3: Backbone Organization and Information Flow

Q3 compares action-only references, separate world/action systems, prediction followed by action, and shared or joint

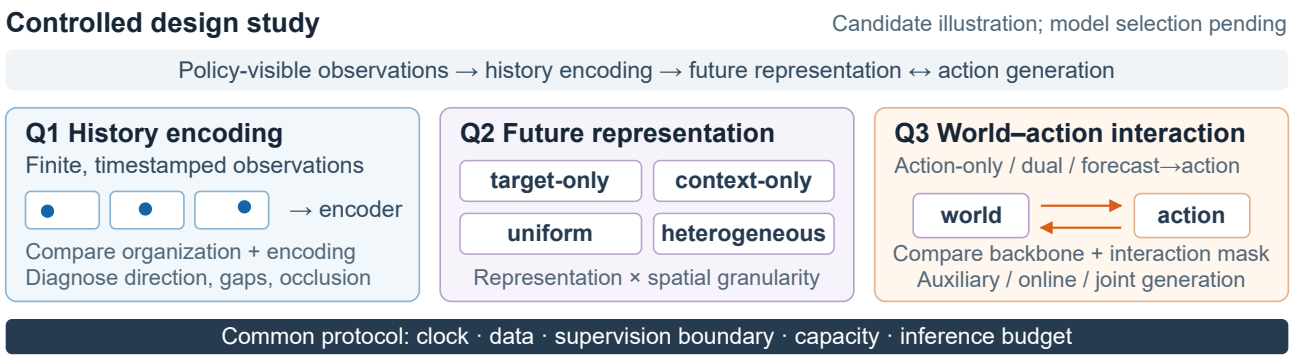


Figure 3. The three-axis design study. Q1 changes the encoding of available history, Q2 changes the future variables and their granularity, and Q3 changes backbone organization and world–action information flow. Common inputs, physical times, and budgets define each comparison. Arrows describe conceptual interfaces; the depicted choices are candidates, not a selected model or measured result.

world–action organizations. Within a compatible backbone, interaction masks determine whether action reads future, future reads action, or both streams interact. Each configuration declares the visible tokens, gradient paths, generation order, and inference cost; a shared name does not make two information budgets identical.

Component controls then distinguish auxiliary prediction, online future conditioning, and joint generation. For the conditional subset these include

$$\begin{aligned} &\text{auxiliary: } p(\boldsymbol{a} \mid \boldsymbol{X}), \quad q(Z \mid \boldsymbol{X}), \\ &\text{online: } p(\boldsymbol{a} \mid \boldsymbol{X}, \hat{Z}), \\ &\text{joint: } p(\boldsymbol{a}, Z \mid \boldsymbol{X}). \end{aligned} \tag{5}$$

Stop-gradient and end-to-end variants address training effects within an applicable online path; they do not replace the backbone study. A compact one-way bridge is one such candidate, with zero, swap, noise, and oracle replacement available when its interface permits a valid intervention. Changing only the future while fixing perception, robot state, and action-source noise tests use; a useful effect also requires appropriate action outcomes and safety.

Inference program versus training path. A forecast-then-act system makes a world output available before its action query. A joint generator may instead update action and world variables together; an action-to-world edge can condition a world output on a proposed action without feeding a corrected future back into that same action. These programs must be described by their actual update order, not just by an attention-mask diagram. The comparison records which intermediate quantities exist when a command is committed and which are available only during training. In a system without an independently replaceable forecast, a bridge swap is inapplicable. In a system that does expose one, a swapped future must preserve the declared interface and reference condition; replacing arbitrary hidden activations would test a different perturbation. Backbone allocation, stopping gradients, and removing an online branch are consequently separate contrasts. Q3 reports their costs and effects without treating any one intervention as a complete study of world–action organization.

Selection and cross-comparisons. Q1, Q2, and Q3 are compared under stated reference choices, retaining competitive alternatives for selected cross-checks. Training-only choices and model selection use the designated development split. The final instance is **[history / future / interaction selection]**; its description is filled only after the study and cross-comparisons. More accurate forecasts, training health, or a fast implementation alone do not select a successful dynamic policy. Detailed losses, normalization, augmentation, and training records accompany the actual candidates in Supplementary Secs. B–E.

5. Experiments

We organize the study around three design questions: **Q1**, how should finite history be encoded; **Q2**, which future variables and prediction granularity help action; and **Q3**, how should world and action computation interact? We first establish a common comparison system, retain competitive Q1 encodings for Q2, and then study Q3 in depth. A small set of cross-checks revisits the provisional choices before selecting a SE(3)-WAM instance. All numerical fields below are pending; the tables specify comparisons rather than measured rankings.

5.1. Common Conditions and Evaluation

The common protocol in Sec. 3 fixes the demonstration split, paired evaluation episodes, sensor channels, action/controller interface, physical clock, and imposed observation–execution delay. Within each comparison we

Table 1. Q1: history encoding under one reference future target and information flow. All values are pending. Win./tok. records physical window and tokens; CTE is contact-time error (ms), and RPE is relative pose error (cm/°). GT object inputs are privileged controls that retain the same EEF history, separate from sensor-only encoders.

History / information	Win./ tok.	SR (%) ↑	CTE ↓	RPE ↓
Strict current-only	0/[K]	[TBD]	[TBD]	[TBD]
Current image + EEF history (H0V)	[TBD]	[TBD]	[TBD]	[TBD]
RGB history	[TBD]	[TBD]	[TBD]	[TBD]
Flow + true time intervals	[TBD]	[TBD]	[TBD]	[TBD]
Sparse point tracks	[TBD]	[TBD]	[TBD]	[TBD]
Estimated object/scene state	[TBD]	[TBD]	[TBD]	[TBD]
Recurrent warped memory	[TBD]	[TBD]	[TBD]	[TBD]
GT current object state	[TBD]	[TBD]	[TBD]	[TBD]
GT direct object history	[TBD]	[TBD]	[TBD]	[TBD]
GT interval kinematics	[TBD]	[TBD]	[TBD]	[TBD]
GT current + kinematics	[TBD]	[TBD]	[TBD]	[TBD]

match initialization, training exposure, optimization, action chunk, parameter envelope, and inference budget; realized parameters, token counts, and computation are reported rather than assumed equal. Motion extraction and future decoding count toward inference cost. We distinguish fresh training from checkpoint adaptation and do not combine them into one ranking. Full configurations and selection rules belong to Supplementary Sec. B.

The primary outcome is task-balanced Core success (SR). We also report task completion (TC), success-conditional completion time (CT), unsafe contact, contact time error, and relative translation/rotation error at contact. Forecast metrics are evaluated at the physical horizons defined in Sec. 3, with representation-specific errors kept distinct from action outcomes. All contrasts use a shared manifest; uncertainty for paired differences is computed over the actual seed and counterfactual-group hierarchy. Sample counts and intervals remain **[pending accepted evaluation release]**, rather than inheriting the historical draft's planned counts. Validation selects references and operating points; Test-OOD and native external tests remain separate held-out comparisons.

5.2. Q1: Encoding History for Dynamic Action

Q1 compares current observations, RGB history, time-scaled flow, sparse tracks, object/scene state, and recurrent warped memory under one reference future target and world–action program (Table 1). The reference is **[history-study configuration]**, not a final architecture choice. Each encoder declares its physical window, sampling times, coordinate frame, visibility, and token budget. Estimated object state uses the same deployable sensor channels; simulator object

Table 2. Q2: future variables and granularity at matched history, reference information flow, and physical horizons. All values are pending. CTE is in ms; prediction errors use declared native units (Supplementary Sec. D), and cost is measured end-to-end p95 ms. Oracle rows do not enter deployable rankings.

Future prediction	SR (%) ↑	CTE ↓	Pred. error ↓	p95 ↓
None (history policy)	[TBD]	[TBD]	–	[TBD]
Target-only pose/trajectory	[TBD]	[TBD]	[TBD]	[TBD]
Context-only motion/geometry	[TBD]	[TBD]	[TBD]	[TBD]
Uniform RGB/latent scene	[TBD]	[TBD]	[TBD]	[TBD]
Uniform flow/point scene	[TBD]	[TBD]	[TBD]	[TBD]
Fine target + coarse context	[TBD]	[TBD]	[TBD]	[TBD]
Motion-selected regions	[TBD]	[TBD]	[TBD]	[TBD]
GT future (matched interface)	[TBD]	[TBD]	0†	[TBD]

†For supplied valid labels only; not an action-error bound.

state and interval kinematics form a separate GT-information block. A current-image condition that retains end-effector (EEF) history is denoted H0V, and is not a strictly memoryless policy.

The main contrast is the strongest matched history policy against its relevant current-input control. Reversed motion with the same current appearance tests direction ambiguity; short occlusions and changed sampling intervals test whether the encoding preserves information unavailable in a single view. History swaps hold the current scene and robot state fixed within valid exact-current groups. We require corresponding changes in contact error and closed-loop outcomes: decodable velocity or action sensitivity alone does not establish useful memory. Results are **[Q1 action/forecast estimates and paired intervals]**. Supplementary Sec. C gives per-task, per-seed, and history-necessity diagnostics. We retain **[competitive encodings]** for Q2 instead of promoting a single action-only ranking to a universal history choice.

5.3. Q2: Future Representation and Granularity

Q2 asks what must be predicted to improve contact position, timing, reachability, and collision avoidance. Table 2 compares no future, target-only, context-only, uniform scene prediction, and heterogeneous prediction using the retained history input and one reference information flow. Target candidates include rigid pose and trajectories/contact points; scene candidates include RGB/latent, flow, and point trajectories. The heterogeneous candidate allocates fine prediction to the target and coarser prediction to relevant context. Static geometry may remain relevant even when it has no motion. This allocation is a hypothesis to test, not the definition of the selected model.

Matching the comparison. Each candidate receives the

same available observation history, supervision episodes, training exposure, physical horizons, and action objective. Region and granularity comparisons use the same declared prediction-token budget where applicable, with actual decoder cost reported. Target-only and context-only conditions remove the other *future* prediction, not its current observation. A whole-scene representation must receive the same task identity and target specification as an object-centric one. Otherwise a gain could reflect extra target information rather than prediction allocation. If a representation requires privileged target state, it is reported in a GT input condition rather than mixed with deployable visual input.

Connecting prediction to action. Native errors describe each predicted quantity; common physical readouts or probes are identified separately when comparing unlike representations. For the rigid-pose/flow candidate, the reserved fields are translation ADE **[X.XX]** cm, symmetry-aware rotation ADE **[X.XX]**°, flow EPE **[X.XX]** cm, and flow–SE(3) consistency **[X.XX]**. These fields do not establish that this candidate should be selected. We relate errors to contact time and relative pose, then to task completion and safety within matched speed, delay, visibility, and motion strata. Lower forecast error can coexist with unchanged or worse action if the corrected variable is irrelevant to the task or costly to compute. The pending comparison is **[prediction–contact–SR relationship, including adverse cases]**.

What supports a role-specific representation? Target-only gains in relative grasp pose or timing would support a target-motion role. Context gains in occlusion recovery, feasible approach, or collision outcomes would support a context role. The orientation contrast uses matched centroid trajectories so that translation cannot explain the difference. If these effects cannot be separated, or uniform prediction matches the heterogeneous candidate at the same budget, we report that result and narrow the allocation claim. Sudden acceleration, stops, piecewise motion, and unexpected impacts receive separate analysis; an unobserved future event is not treated as a deterministic forecast target. Prediction after robot contact is scored only under the corresponding applied action conditions.

Oracle headroom. Future oracles provide ground-truth variables at the same horizons while keeping the action interface and remaining inputs fixed. They are privileged references, separate from GT *current/history* inputs. For a predicted-future policy P, the strongest matched history policy B, and an interface-matched future oracle O, we reserve the oracle-gap fraction

$$G_{\text{oracle}} = \frac{\text{SR}(P) - \text{SR}(B)}{\text{SR}(O) - \text{SR}(B)}. \tag{6}$$

Its pending value is **[XX]**%; it is interpretable only with a supported, non-negligible denominator. Oracle access is not evidence of visual localization or deployable forecast accuracy. Supplementary Sec. D reports full horizon/region sweeps, calibration, and negative results.

Table 3. Q3: backbone/program comparison with provisional history and future targets fixed. W and A denote world and action; arrows indicate inference information paths, not gradient paths. All outcomes and p95 loop cost (ms) remain pending; unsafe contact is a percentage.

Backbone / inference program	Online path	SR (%) ↑	Unsafe ↓	p95 ↓
Single action reference	A	**[TBD]**	**[TBD]**	**[TBD]**
Separate world/action systems	W ‖ A	**[TBD]**	**[TBD]**	**[TBD]**
Forecast-then-act (FTA)	W → A	**[TBD]**	**[TBD]**	**[TBD]**
Coupled, action reads world	W → A	**[TBD]**	**[TBD]**	**[TBD]**
Coupled, world reads action	A → W	**[TBD]**	**[TBD]**	**[TBD]**
Bidirectional joint generation	W ↔ A	**[TBD]**	**[TBD]**	**[TBD]**

5.4. Q3: World–Action Backbone and Information Flow

With provisional history and future targets fixed, Q3 compares an action-only reference, separate world/action systems, forecast-then-act (FTA), and coupled backbones with world-to-action, action-to-world, or bidirectional paths (Table 3). These are system comparisons: parameter allocation, training updates, visible variables, gradient paths, decoding order, and execution cost must be stated for each. A preliminary backbone screen establishes a usable reference for Q1–Q2; it does not replace this deeper study or establish the final winner. Results remain **[Q3 system estimates and paired intervals]**.

We test whether action benefits from reading world information and whether the reverse path adds value under the same end-to-end budget. The direction of an attention or conditioning path is distinct from the direction of its training gradient. In particular, an action-to-world path can improve world prediction without supplying a future to the same action decision. Any benefit must therefore be explained using the actual inference program rather than a generic claim of joint modeling.

Component diagnostics then separate future auxiliary supervision, online forecast consumption, and joint generation within compatible systems. Disabling an online path tests its deployment contribution while preserving the trained model; retraining without future supervision tests a different training effect. For an explicit replaceable future interface, exact-current interventions reserve swap action change **[X.XXX]**, swap SR change **[-X.X]** percentage points, and zero/noise SR changes **[-X.X]**/**[-X.X]** points. Zero, swap, and noise are used only where the manipulated interface exists and the intervention is meaningful. A system with no online future path has no corresponding intervention, rather than a measured zero effect. Useful forecast consumption requires

agreement among behavior, safety, forecast quality, and oracle direction; sensitivity alone is insufficient. Supplementary Sec. E gives these diagnostics, the matched auxiliary controls, and full system configurations.

5.5. Cross-Checks, Selected Instance, and Cost

The final choice follows two targeted cross-checks: repeat retained Q1 encodings after changing Q2's future representation, then repeat retained Q2 representations after changing Q3's information flow. We report **[ranking stability or reversals and affected conditions]**; a reversal triggers a revised conditional choice rather than removal of the inconvenient comparison. Supplementary Sec. E lists the retained combinations. Data/model-scale checks subsequently test whether the selected comparisons persist; they are not substitutes for the matched small-system study.

The selected instance is **[Q1 encoding + Q2 representation + Q3 program]**. Its reserved Core fields are SR **[XX.X]**%, TC **[X.XXX]**, CT **[X.XX]** s, and unsafe contact **[X.X]**%; the strongest matched history baseline has SR **[XX.X]**%, giving **[+X.X]** percentage points (95% CI [**[X.X, X.X]**]). These fields refer to the eventual selected instance, without assigning it in advance to P-Detach or another candidate.

Imposed delay is a controlled task condition; measured execution cost is a separate result. On **[hardware and batch-one profiling setup]**, the reserved end-to-end p50/p95 loop times are **[XX.X]**/**[XX.X]** ms, with p95 action age **[XX.X]** ms and deadline misses **[X.X]**%. Supplementary Sec. F decomposes preprocessing, history encoding, world/action computation, memory, and scheduling. Native external/static protocols and real conveyor evidence remain **[TBD]** (Supplementary Sec. F), with no transfer or real-robot claim filled in yet. Failure cases that change a design conclusion stay in the main comparison; Supplementary Sec. G aligns further predictions, observations, rollouts, and anonymous video files in physical time.

6. Limitations and Conclusion

This study examines how history encoding, future representation, and world–action information flow affect dynamic manipulation under matched observation, training, and execution conditions. Its outcome will be a condition-specific design choice, **[supported Q1–Q3 findings and selected instance]**; numerical results and external validation remain pending. The current scope is short-horizon, single-arm manipulation of task-specified rigid objects. It does not establish language-based target selection, long-horizon planning, deformable-object control, or unseen action-conditioned contact dynamics. Unannounced events require new observations, and forecast confidence alone provides no safety guarantee. The final claims must retain any negative results and interactions that alter the design choice, rather than presuppose that each study axis improves performance.

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